Name _____ _____ Baggett/Fall2026/207/Quiz6

first name last name

Score

No Calculators. / 20

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 1 for the reason.**

1. (**2** pts) In a related rates problem you differentiate the relating equation with respect to time *before* substituting the given numerical values.

True **False**

Why? _____ /2

2. (**2** pts) $\frac{d}{dx}(\ln 10) = \frac{1}{10}$

True **False**

Why? _____ /2

3. (**6** pts) **Derive** $\frac{d}{dx}[\arcsin x]$ using implicit differentiation. **Show every step.**

You may use the derivative rules for sin and cos, but **not** any derivative rule for an inverse trig function. Quoting the formula earns nothing.

_____ /6

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4. (**3** pts) Find $\dfrac{dy}{dx}$ by implicit differentiation:

$$x \sin y = y$$

/3

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5. (**2** pts) Differentiate $y = 5^{x^2}$. You do not need to simplify.

/2

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6. (**3** pts) Use **logarithmic differentiation** to find $\dfrac{dy}{dx}$ for

$$y = x^{2x}, \qquad x > 0.$$

/3

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7. (**2** pts) Each edge of a cube is growing at 2 cm/s. Let s be the edge length and V the volume.

Write the relating equation and **differentiate both sides with respect to t. Do not substitute any numbers and do not solve.**

/2

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